

COVID-19 Vaccination Data Visualization: Issues and Challenges

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Abstract— Data Visualization plays an important role for patterns and trends analysis in trillion of data rows Big Data analysis, where the data can be represented in some graphical forms. Hence, the data could be more comprehensible in its visual summary in dashboards and storyboards. This study aims to discuss some issues and challenges in visualizing COVID-19 vaccination datasets. There are some possible issues in data visualization, as it is not easy and may be challenging to produce a good dashboard that are interesting and easy for viewers to understand. Therefore, this study focuses on some issues that may arise during performing a data visualization on the COVID-19 dataset. **In this study, there are three dashboards have been studied, which are the COVID-19 tracker, its effectiveness, and its acceptance. The first two dataset are derived from Ministry of Health Malaysia bank data, whereas the third dataset is from a survey to support this analysis.** The selected attributes are states, the number of people who have received the vaccine as adults, children, and teenagers, and the number of people who already received boosters, and reasons to not get a booster. The visualization issues found within the dashboard are mis-choice of colors, mis-choice of visual object type, lack of interactivity, and plotting too much data. As a result, this proposed alternative solutions for those issues such as color deliberately, pick a suitable visual object, create an interactive dashboard, and reduce the information overload in visualizing the data.

Keywords—Data visualization; COVID-19 Vaccination; Data Analytic; Big Data Analysis; Interactive Dashboard.

I. INTRODUCTION

As the age of Big Data gets more spearheading, visualization has become an increasingly crucial tool to make sense of the trillions of rows of data generated every day. By definition, data visualization is the graphical representation of information and data using visual elements like charts, graphs or any data visualization tools to provide an accessible way to see and understand trends, outliers, and patterns in data [1]. Data visualization has essential uses across several factors, from science and technology to finance to education, and presently, data visualization is comprehensively used in the medical and health sectors, especially after the COVID-19 outbreaks. Currently, the healthcare sector is also no exception in joining the Big Data to collect and track mountains of

information regarding the disease spread or health statistics. Even though a lot of data is available from many sources, it is impractical for the medical staff to analyze the data without data visualization to find required information faster and fasten the decision-making process. Modern data visualization tools in healthcare transform complex data into user-friendly visuals that are very easy to understand for its stakeholders, whether they are doctors, patients, or government officers. This health-related data is also very much needed by the public to know current trends and issues, especially during this COVID-19 outbreak.

Since 2020, the coronavirus SARS-Sov-2 (COVID-19) has been declared by the World Health Organization (WHO) as a pandemic [2]. There are a lot of prevention steps guided by the WHO to minimize the severity of this pandemic, including formulating various vaccines to strengthen human immunity to fight the COVID-19 virus. In correlation with the initiative suggested by WHO, on 6th October 2021, the Malaysian government made the vaccination mandatory for Malaysians to reduce the death rate caused by COVID-19 in Malaysia [3]. As the vaccination drive against this virus in Malaysia progresses rapidly, researchers have conducted many findings to examine the vaccination progress and its effectiveness. Hence, there is a need for data visualization to display real-time data, showing patterns and trends over time and also more advanced analytics such as figuring out why some people refuse to get vaccinated [4].

In Malaysia, the Ministry of Health (MoH) has collaborated with the open data community to launch an interactive dashboard known as COVIDNOW. The main purposes of the COVIDNOW dashboard are to share fresh insights and analysis of the situation and help Malaysians to monitor the epidemic. However, constructing a data visualization dashboard is not an easy task, as implementing and using data visualization also quite challenging. This study attempts to analyze some major issues and problems in data visualization especially in visualizing vaccination progress and effectiveness and propose some alternative solutions to handle those issues.

II. RELATED WORK

A. Data Analytics and COVID-19 Vaccination Analysis

Presently, the gathering, storing, manipulating, and representation of big data has surged immensely, especially with the help of social media and mobile device usage [5]. Seeing that the technology era is becoming more modern day by day, the continued advancements of software programs combined with top-of-the-range hardware development has allowed big data to be used and presented in more advanced ways than before. The commonality of the usage of charts and graphs has now been allowed to be utilized in superior ways than before to represent the data, presented in more colorful ways, making information clear and easier to understand [6]. Correspondingly, data visualization has become one of the most conventional and crucial elements in interpreting and presenting data to the audience since organizations also have begun to emphasize gathering and analyzing large data [5].

Big data trend has been used extensively in many areas, such as predicting election results, stock performance, and trends within health care [7]. It is apparent that the requirements of big data are becoming more significant and are needed to pour the necessary research, attention, and knowledge to gain insights into big data analytics. In spite of that, utilizing big data requires a lot of energy to offer visible data to the eyes and maintain and improve performance as advancements in data analytics continue [5]. Hence, it is required to study the importance of the methods used in analyzing data and how they can assist in determining different data patterns that may not be visible initially. In pandemic stage, data visualization can be exploited to go through different fallouts that have occurred during COVID-19 and the fallouts that will continue once COVID-19 is over. There are so many ways in which the data can be interpreted, such as using the data to create a larger vision or breaking down the information given using various descriptive and exploratory methods [6]. Furthermore, with the effects of social media, it is essential to establish trends and interpret and analyze the data so the data can be spread fleetingly worldwide.

B. Dashboard Development for Real-Time Visualization of COVID-19 Vaccine Safety Surveillance Data in the Vaccine Safety Datalink

Dashboards could quickly identify trends and irregularities by summarizing the most valuable information from enormous amounts of data. To end this COVID-19 pandemic, vaccines are the crucial element of public health [8]. However, the safety of the vaccine is an important part of it since it involves people's health. Therefore, it is important to develop a near real-time visualization to track the COVID-19 vaccine safety for the population. Throughout the COVID-19 pandemic, the dashboard could benefit health care by tracking the COVID-19 cases and deaths and tracking the progress of vaccination among the population. Kenigsberg et al. [8] used a variety of COVID-19 vaccine Rapid Cycle Analysis (RPA) and COVID-19 disease monitoring variables to produce the key metrics monitored in the dashboard. Based on the key metrics, it served as the basis for dashboard content, which enables the users to quickly review large amounts of data and easily share visualizations. The authors mentioned that the developers of the

dashboard displayed seven major categories in the Vaccine Safety Datalink (VSD) COVID-19 Vaccine Dashboard, which contained COVID-19 vaccine administration; COVID-19 vaccine coverage; descriptive summaries of outcomes following COVID-19 vaccination; statistical analyses of COVID-19 vaccine outcomes; COVID-19 disease hospitalization counts; descriptive summaries of simultaneous vaccine administration with COVID-19 vaccine; and descriptive summaries of COVID-19 disease and COVID-19 vaccination during pregnancy. According to the authors, the dashboard creators used a tool known as Tableau Software to create the VSD COVID-19 Vaccine Dashboard. This platform allows for flexibility in both design and content and the with visualization is regularly updated to keep up with the evolving COVID-19 vaccination landscape. In addition, the arrangement and design of these visualizations are constantly being improved by the developers, ensuring that the contents of the dashboard remain relevant throughout the year. Therefore, the dashboards for vaccine monitoring data are helpful tools for rapidly creating visualizations to support data exploration in public health sector and could benefit the government in quickly evaluating and responding to critical question regarding vaccination safety [8].

C. Interactive Visualization of Large Data Sets

Usually, an interactive visualization is used so the viewers can understand a subset or aggregation of the data, through selection or projection. Analyzing large data sets are easier if it is facilitated by interactive data visualization. Therefore, the user can select, navigate, and display the data through user friendly interface [9]. It gives the user the flexibility to switch views, make filter selection, and further understand the trend of the data [10]. According to [10], interactivity in data visualization comes from Bertin, who suggested that ability to interact with visual representations is a sign of a good visualization. According to [11], "cognitive artifact" is important, and it means that the user could interact with the large data set using filter and slicer tools. There are a few tools that offer interactive data visualization. These include CHART, APT, SAGE, InfoCrystal, IDEs, VisDB, VIS, IVEE, Visage, Tioga-2, Spotfire, DataSpace, DEVise, DataSplash, Rivet, Polaris, and Tableau. Besides, the first tools to add interactivity in visualization are CHART and BOZ where manipulation could be done to make interactive chart. In modern visualization system, the tools put more efforts into making interactive visual objects. For example, Rivet makes it possible to make one visualization having multiple views of the same data. The next modern visualization tool is Polaris which provides deriving additional fields, selecting details, sorting, filtering, brushing, zooming in-out, and tool tips. These functionalities are also made available for the next visualization tool that comes after, Tableau which inspired by Polaris [12]. There is also POWER BI which developed by Microsoft [10]. In modern tools, the integrated database is also stored in a table-like form, so visualization becomes faster.

The challenge in data visualization identified by the author is the visual scalability. In other word, large data must be compressed or reduced before it could be visualized. Therefore, the suggestions to address this problem are to filter only interested subset of data or attributes for visualization, to use

“group by” clause in SQL query to aggregate some numeric data into sum or average, and to sample the data so that only certain data rows are used for visualization. Although, it could also be suggested that interactivity may also overcome the crowded screen of data points, so user may select which category to be viewed.

D. Guidance on Data Visualization

It is possible to define data visualization as the process of putting information into a visual context, such as a map or graph, to make it simpler for the human brain to comprehend and draw conclusions from the data [13]. The main goal of data visualization is to make it easier to identify patterns, trends, and outliers in large data sets. Therefore, data analysts and researchers can follow several pieces of advice to create exquisite yet knowledgeable visual objects. The use of color should be carefully strategized to deliver the key findings [14]. Most data should be in neutral colors like grey. It is best to avoid using rainbows and vivid colors in constructing visual objects. Bright colors should be reserved to direct attention to significant or atypical data points. The author also advised to use the color when they should, not when they can.

Another key point of a good data visualization is the excellent choice of the visual objects. For example, the horizontal bar chart is used when graphing nominal or categorical variables [15][16]. Moreover, a horizontal bar chart is a better choice for long category names because more space on the left-hand side. In contrast, the vertical bar chart is suitable for numerical such as ordinal and sequential data [15]. In a vertical bar chart, labels might overlap, and would need to be rotated or shifted to be readable [17].

III. DATA VISUALIZATION ISSUES

There are four possible issues are identified for data visualization for the dataset. The issues are mis choice of color, choosing the wrong visual object, non-interactive chart, and data plotting. In the next subsections, more detailed explanation on each issue would be discussed.

A. Selection of Color

Color plays a vital role in data visualization as it is among the most persuasive design elements. In data visualization, high degrees of color contrast may cause viewers to believe those value disparities are greater than they are [18]. Color is one of the most common and effective ways to deliver information, allowing data analysts to highlight certain pieces of information. One of the frequent mistakes in communicating data using color is encoding too many data points in a different color. The color palette is huge, which can lead the designers to use too many or too few colors. Colors can be an incredibly effective tool to enhance data visualizations. However, poor color choices can ruin an otherwise excellent visualization [19]. Coloring in data visualization must not be done for the sake of coloring only; each color should have a clear purpose. As an example, fig. 1 shows a poor color implementation in data visualization.

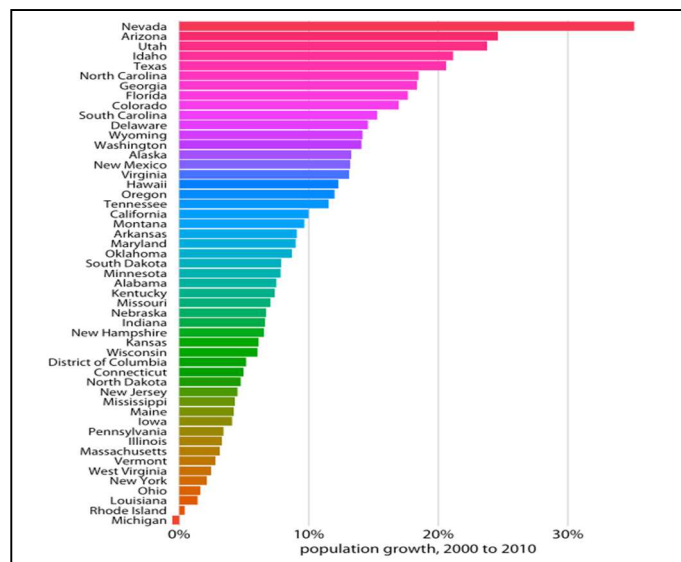


Fig. 1. Example of aimless use of color [19]

In fig. 1, each bar is given its color, and the bars create a rainbow effect. Although it may give visual impact to this chart, it does not create any new insight into the data or make the figure easier to read. Besides that, fig. 1 has a second color-related problem; the chosen colors are too saturated and intense, which makes the figure difficult to look at. It is also difficult to read states' names without readers' eyes drawn to the large colored areas right next to the state names. It is difficult to compare the endpoints of the bars and focus on which state has the highest number of populations. The purpose of this case study which is to identify the issue in visualizing the vaccination progress and effectiveness in Malaysia, there is an example that are related to this topic. Fig. 2 shows a graph that represents vaccination doses administered in several countries.

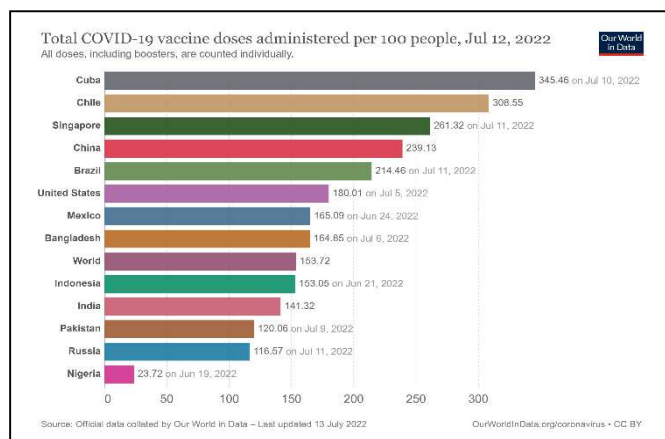


Fig. 2. Total COVID-19 vaccine doses administered per 100 people in some countries [20]

This graph may be hard to be interpreted by the readers who may have some form of color-vision deficiency such as colorblind. These readers may not be able to distinguish colors that look clearly different to most other people. Approximately 8% of males and 0.5% females suffer from some sort-of vision deficiency. Vision deficiency has some variants such as people

who have difficulty to perceive green or red is known as deuteranomaly and protanomaly while people who cannot perceive the blue color is known as tritanomaly [19]. In short, using multiple colors, bright and concentrated without any purpose is the most common issue in data visualization in terms of color aspects.

B. Visual Objects Selection

There are numerous visual objects that data visualizer could choose from for visualization. However, it should be noted that each visual object should be matching with their functionality and purpose. Fig. 3 shows different kind of visual objects and their respective purpose. For example, scatter plots and pie chart are to show distribution and composition, respectively. In the other hand, bar chart has more than one purpose, which are whether to show distribution, comparison, or composition.

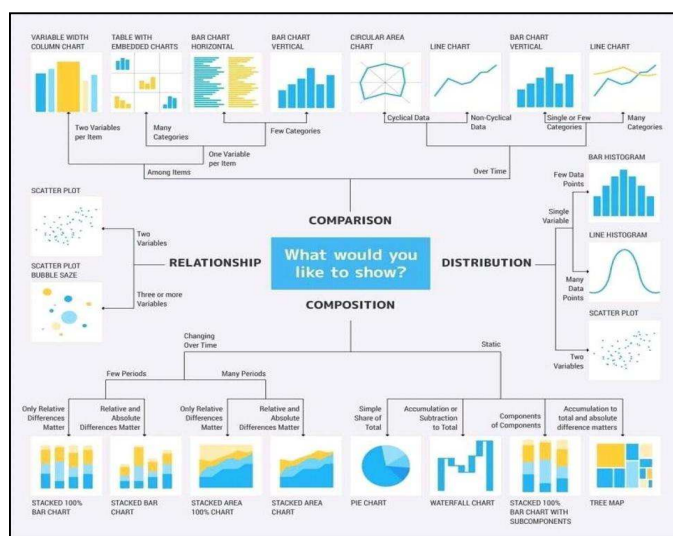


Fig 3. Functionality of different visual objects [21]

Another example, the focus is the second dataset, which is the survey. In the second dataset, there is an attribute to be visualized, which is “Reason People Not Getting Their Booster.” The data from this attribute was presented in the form of tree map in this dashboard.

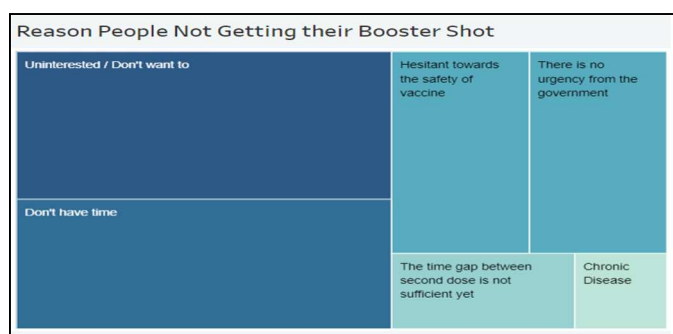


Fig. 4. Tree Map of Distribution Reasons People Not Getting their Booster Shots [22]

The problem with this visual object is the tree map has been misused. To properly use a tree map, the data must be hierarchical in the sense that it could be separated into subcategory. This issue is because tree maps are utilized to

display data that is grouped and nested in hierarchical order [23]. Fig. 5 show the correct way to use tree map. The categorical data is represented correctly in their rectangles. Comparing to the tree map of the dashboard earlier, the reason why people did not get their booster shot do not have any clear subcategory.

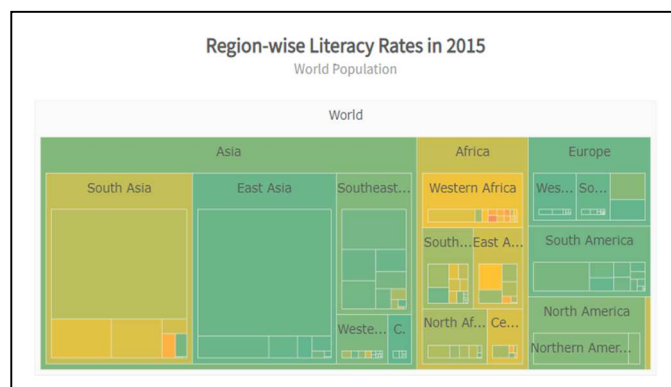


Fig. 5. Tree Map of Region-wise Literacy Rates in 2015 [23]

As an example, in the dataset, “Chronic Disease” does not have subcategories such as “myocarditis” or “chronic kidney disease”. Tree maps are excellent chart for displaying data such best-selling products, location population, regional sales, and similar parent-child structured data [24]. The ideal use of tree maps is also when multiple data categories are large, and with space constraints [23]. In conclusion, the use of tree map in fig. 4 is not suitable and need to be replaced with another visual object.

C. Non-Interactive Chart in Dashboard

A non-interactive chart is one that does not incorporate any interaction capabilities and does not change with time. Non interactive chart in data visualization is a chart that are static and simple [25]. Static visualizations are commonly seen as infographics posted on the web where users cannot go beyond a single view to explore additional stories beyond what’s in front of them. User could have wrong ideas about the concept of certain issues. This does not mean that non-interactive chart can’t be used in data visualization. But interactive chart is more efficient to uncover the insights which lead to better, data driven decision since it allow users to engage with the data and easily find the answer to the user specific queries. However, a non-interactive visualization is more suitable for less complex data stories which only perform one or a few queries.

Fig. 6 below shows the example of non-interactive chart in the dashboard. From the dashboard below, the graphs is a non-interactive chart. User could not filter the data by the state. Due to lack of interactivity between the user and the dashboard, these serve a limited purpose since the user can’t explore beyond the data. A picture of graph, or any other form of static visualization will not only fail to convince a user of the inconsistencies in their views, but it will also be perceived within the context of their assumption.

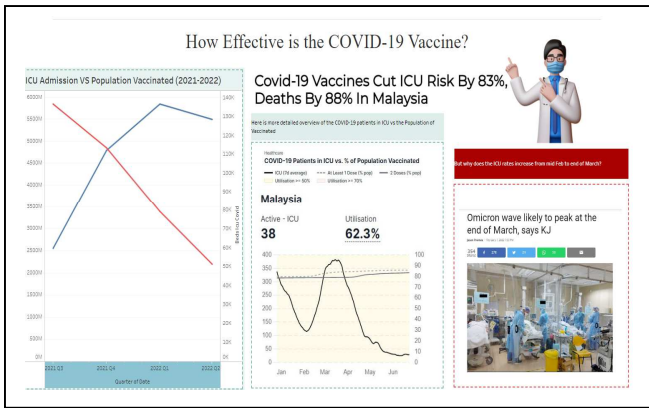


Fig. 6. Example of non-interactive dashboard [22]

D. Plotting Too Much Data

Entering numerous data points into a single chart forces readers to process heaps of data at a single glance, making the value of such training far from clear [26]. Sometimes, without realizing it, users plot so much data that it becomes impossible to see the detailed shapes of the graph [27]. Fig. 7 shows a chart that has too many data points to count. Cramming too much information in a single graphic is impractical and prevents users from successfully interpreting the data presented. Visualizing data for the sake of visualization is counterproductive and unhelpful to the readers. The visualization in the fig. 7 below fails to compare ATM adoption rate by country effectively even though it is the main purpose of the chart. The sheer number of data points in this chart is overwhelming and can cause the readers to struggle to derive the message that this chart attempted to convey [26]. Another example of cramming too much information is shown in the fig. 8 below.

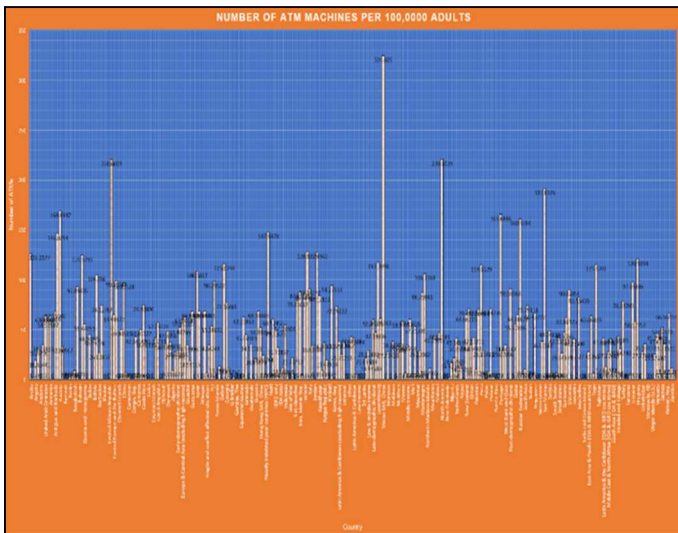


Fig. 7. This chart contains too many data plotted on it [28]

In fig. 8, the chart tries to visualize the vaccination progress by state categorized by age. Even though this chart clearly shows the vaccinated population figures by states, there is so much information to be digested by the viewers at once. Hence, it is also considered as a severe issue in data visualization.

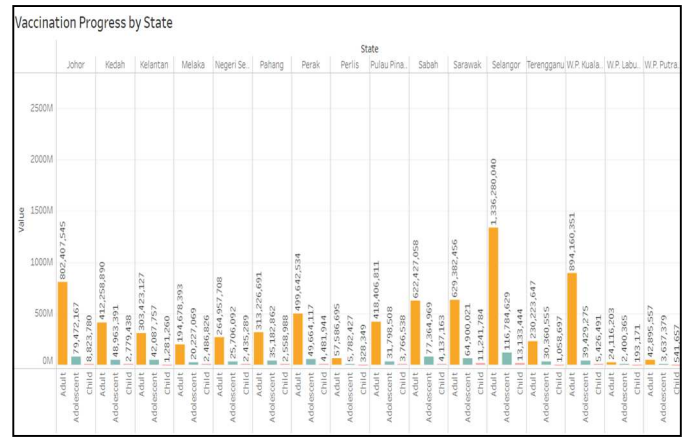


Fig. 8. Another example of cramming too many data in a chart [22]

IV. RESEARCH METHOD

In this case study, the dataset source is from the Ministry of Health Malaysia. The health ministry has published open data on the COVID-19 pandemic in Malaysia for the public to view. The dataset name is Open Data on COVID-19 in Malaysia, which is available on the GitHub platform (<https://github.com/MoH-Malaysia/covid19-public>). This dataset tracks the daily and total vaccination and ICU admissions due to COVID-19 in Malaysia. The dataset comprised two major areas, which are information about vaccination and ICU that related to COVID-19. Therefore, to further understand the data, it was observed under the 5 v's of big data, which are volume, velocity, variety, veracity, and value. The volume of the data is 7665 total rows. The velocity of the data is that it was originally generated by MySejahtera. MySejahtera is an application developed by the Government of Malaysia to assist in managing the COVID-19 outbreaks in the country. Following that, the variety of this dataset is not complex meaning that the data is structured data and has been organized into a formatted repository, means the data is made more accessible for effective analysis. Next, the veracity of the dataset is accurate and trustworthy since the data is clean and it is from the ministry of health in Malaysia. Finally, for value, the value of the dataset is interesting because it can be useful for health and academic researchers or may assist the government in advocating and elevating general populations' willingness to get the COVID-19 vaccine. Table I summarizes the data column contents and data description for the vaccination and ICU admission.

TABLE I. DATASET DETAILS OF OPEN DATA ON COVID-19

Data Column	Data Description
state	Name of 13 states in Malaysia
cumul_full	Total number of people who have completed their vaccination regimen.
cumul_full_adol	Number of adolescent (12 to 17 years old) who have receive two doses of vaccine.
cumul_full_child	Number of children (5 to 11 years old) who have receive two doses of vaccine.
bed_icu_covid	Number of ICU admission into the hospital due to COVID-19

The second dataset source comes from the online questionnaire on vaccine recipients. This dataset describes the online survey given to the respondent about the vaccine booster.

A total of 98 responses were collected for this online survey. In this questionnaire, respondents were mostly asked about the booster shot and the reason for still not getting it. Table II below shows the contents of the data columns and data descriptions for the online questionnaire dataset. Fig. 9 shows the sample of dataset from the survey questions. It displays the answer for the reasons of the respondents still not getting the booster shot.

TABLE II. DATASET DETAILS OF THE ONLINE QUESTIONNAIRE ON VACCINE RECIPIENTS AND EFFECTIVENESS

Data Column	Description
reason_not_booster	The reasons of the respondents still not getting booster shot

Why are you still not getting a vaccine booster?
Chronic Disease
Uninterested / Don't want to
Hesitant towards the safety of vaccine
Don't have time
There is no urgency from the government
It is not mandatory except for those who take sinovac vaccine
The time gap between second dose is not sufficient yet

Fig. 9. Sample of dataset from the survey questions

V. RESULT AND DISCUSSION

A. Choose Color Carefully and Deliberately

As mentioned in the previously, color plays a vital role in data visualization as it is among the most persuasive design elements. The human eye is drawn to pattern and colors and their brain can process and take in necessary information better if the spreadsheets and reports is presented in a visual manner. A common mistake inexperienced data analysts tend to make is to integrate either too many or too few colors in the visual objects. They are supposed to choose color carefully and deliberately. Like all stylistic elements in a visualization, selecting colors should not be a purposeless choice. Color is important to highlight information, illustrate a steady or a sudden change, as well as helping to visually compare or contrast data points. As discussed in fig. 1 and fig. 2 earlier, each bar is given its color, and the bars create a rainbow effect. Not only that, but the graphs may also be difficult to be interpreted for someone who is visually impaired to understand the COVID-19 vaccine's progress and effectiveness. Fig. 10 shows a suggestion on how to show the COVID-19 Progress in Malaysia by States using graph that implements the correct technique of choosing the colors to tackle these issues. In fig. 10, all the bars are only colored in grey, except one bar is colored in green to show which state has the highest number of vaccinated people. This graph looks more aesthetically appealing to viewers than the graphs in fig. 1 and 2. The viewers can also recognize which state has the highest vaccinated population in an instance and helps people with color-vision deficiency. The future and current data analysts should put in their minds that color can be an incredibly effective tool to improve data visualizations. Therefore, color must be applied to serve a purpose clearly and must not distract from the main objective.

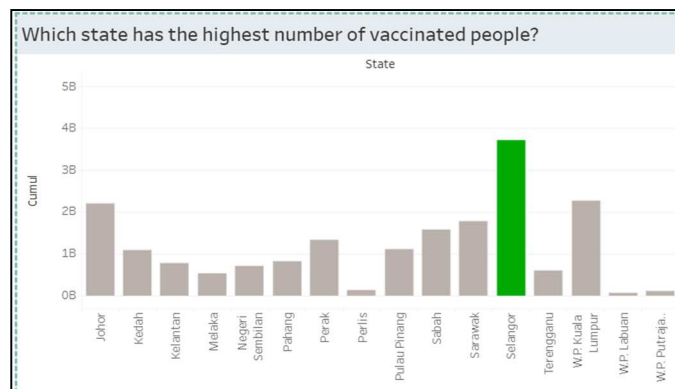


Fig. 10. Suggestion on the COVID-19 Progress in Malaysia by States

B. Pick a Suitable Visual Object to Represent the Data

To recap, Scenario B. is the wrong usage of the tree map to represent the reasons why people do not get their booster. Therefore, to solve the problem, the purpose of the visualization must be evaluated and truly determined. In this case, the purpose of visualization is to show the comparison of the number of people for different reasons, which are “Uninterested”, “Don’t Have Time”, “Hesitant towards the Safety of the Vaccine (Booster)”, “There is no Urgency from the Government”, “The Time Gap between Second Dose is not Sufficient yet”, and “Chronic Disease”. These reasons are considered one variable under “Reason”. It does not have subcategory, or different variable to be compared. For showing the comparison of the number of people in different categories of reasoning, with one variable per category, the most obvious reason would be the bar chart. There are two types of bar charts in consideration, which are horizontal and vertical bar charts.

Hence, based on the guidelines in similar work, the most suitable chart to represent the “Reason Why People Not Getting their Booster Shots” is horizontal bar chart. This is because not only the axis is nominal, but the category names are also long. For example, “Hesitant towards the Safety of the Vaccine.” Fig. 11 is an example of a good horizontal bar chart, produced by Ministry of Health to compare COVID-19 cases in different states. In conclusion, to have a good data visualization, one must determine the purpose of the visualization and decide which chart is best suited to the purpose.

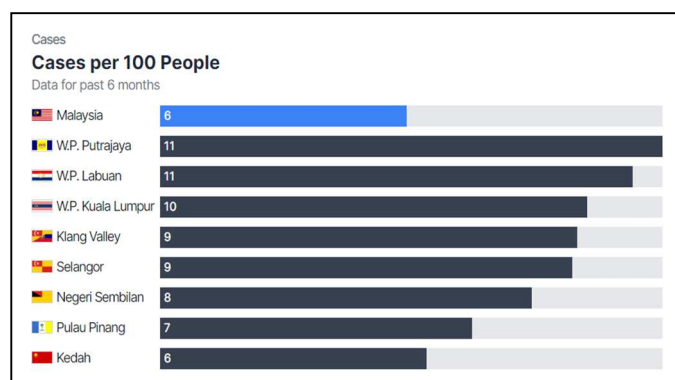


Fig. 11. Example of Horizontal Bar Chart used to Visualize COVID-19 Cases

C. Create an Interactive Dashboard

An interactive dashboard also plays a vital role in data visualization since it enables users to act on the data instantly to easily generate and filter reports to understand the underlying issues [29]. Interactive visualization allows users to engage with data in ways not possible with static graphs. Therefore, it is recommended to provide interactive visualization in the dashboard to resolve issues with the non-interactive chart. For visualizations to be considered interactive, it must involve an aspect of human input, such as clicking on a button or adjusting a slider, and the response time must be fast enough to show a real relationship between the data input and visual output [30]. In addition, interactivity is the best option for enormous amounts of data with complicated data stories since it provides the ability to discover, separate, and visualize information for extended period of times [25]. Filtering, zooming, and panning are common features to apply in interactive visualizations.

Fig. 12 below displays the dashboard with the filter option. Filtering allows users to reduce or specify the data on the dashboard as temporary filter values. From the dashboard below, users can filter the data by selecting the specific state to view the trends between the COVID-19 patients in ICU and overall vaccination rate. In addition, users can simply return to an unfiltered view by using the “Remove all” filter, or just click into another state to access that data. With only a few clicks, users can easily interact with data and discover what is related to them.

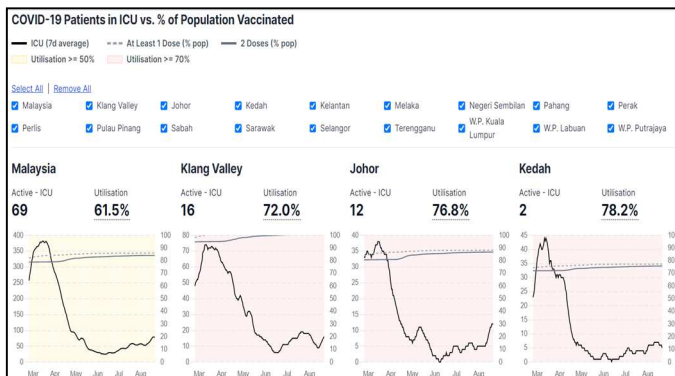


Fig. 12. Example of interactive dashboard with a filter option

Another feature that can be used in creating an interactive dashboard is, zooming and panning. Zooming and panning are useful if a user wants to see a specific detail in the chart. A user can zoom in and focus on a specific part of the representation without being distracted or needing to generate a new insight.

D. Reduce Information Overload in Visualizing

Finally, yet importantly, do not plot too much data in a chart. When planning data, it is important to stick to a specific aim. Otherwise, the audience cannot follow its meaning or find a clear takeaway and ignore the content that is tried to convey in the data visualization. Some researchers suggest that people usually spend only an average of 5.6 seconds on a website's content [26]. As mentioned previously, cramming too much information prevents the audience from successfully

interpreting the data presented. As mentioned before in Fig. 7 and 8 have shown how intricate when a lot of data is loaded into one chart. To avoid this issue, it might be better to separate the information into several charts, as shown in Fig. 13 and Fig. 14 respectively.

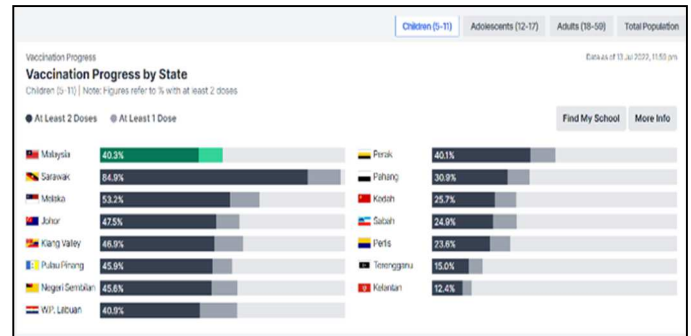


Fig. 13. Vaccination Progress by State among Children aged 5 to 11 years old

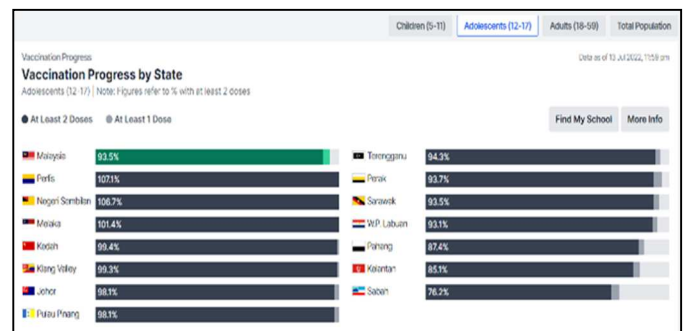


Fig. 14. Vaccination Progress by State among Adolescents aged 12 to 17 years old

Those analysis are a good example of organizing and planning a lot of information from large dataset. These figures are separated by age categories using interactive buttons shown in Fig 15.



Fig. 15. Interactive buttons to separate vaccination progress charts

It is a good approach by the MoH developers' team to use a percentage instead of using the exact number figures to show the number of individuals. Not only that, but the audience can also easily understand other information details such as the total population that has received at least one dose or two doses and see which state has the highest and the lowest vaccinated people by age categories. These figures immediately capture the viewer's attention and give them the information they want as they can compress all the data into limited spaces without looking too cramped. There are some suggestions on how to achieve a good visualization when delivering a lot of data. Being able to convey the most exciting data will immediately rouse people's interests and attract more people by engaging their gaze and drawing more positive information to the visual objects presented [31]. The data analysts must remember to simplify the data, improve readability by leaving only the essential details and find a way to allow the viewers to link the

information with engaging visual objects that can make the data more presentable [31].

VI. CONCLUSION

There are four issues that are highlighted in data visualization for related to COVID-19 Vaccination datasets, which are issues on the selection of colors, the selection of charts, the interactivity of the visual objects and dashboard, and the handling overloaded of information in data visualization. In this study, those issues have been addressed and discussed. Then, this study proposes that alternative solutions to handle those issues. This issue may be related to the knowledge of the data analyst, or the tools used. However, since the functionality of the visualization tools such as Tableau and Power BI are quite similar and have their own pros and cons, it is ambiguous to pinpoint the specific functionality of the tools to relate to the solution based on the problem. Nevertheless, as important as data visualization is to further understand the data trends, it is also very critical to understand the method of applying it correctly. The failure to do so may affect the dashboard appeal and the storytelling to the audience. A good dashboard and storyboard must apply a good color choice on the visual object, use a proper chart to represent the data, have a good amount of interactivity, and proper plan which information to be placed to avoid overcrowding the storyboard. Therefore, this case study's aim is to address these issues and to lay out the solutions that have been achieved.

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